Algorithm Name

AMSGrad with bias correction and monotone second-moment accumulator

Mathematical Formulation

We minimize a (possibly non-convex) differentiable objective $f : \mathbb{R}^d \to \mathbb{R}$. At iteration $t \geq 1$, we observe a (possibly stochastic) gradient $g_t \in \mathbb{R}^d$. Elementwise operations are denoted by $\odot$ (product), $\oslash$ (division), and powers are taken elementwise. Let hyperparameters - base stepsizes $\{\eta_t\}_{t\geq 1}$, - momentum $\beta_1 \in [0,1)$, - second-moment decay $\beta_2 \in [0,1)$, - numerical stability $\epsilon > 0$.

Define for $t \geq 1$:

$$\begin{aligned}
m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\
v_t &= \beta_2 v_{t-1} + (1-\beta_2)(g_t \odot g_t), \\
\hat{m}_t &= \frac{m_t}{1-\beta_1^t}, \qquad \tilde{v}_t = \frac{v_t}{1-\beta_2^t}, \\
\hat{v}_t &= \max(\hat{v}_{t-1},\, \tilde{v}_t) \quad \text{(max taken elementwise)}, \\
w_{t+1} &= w_t - \eta_t\, \hat{m}_t \oslash \big(\sqrt{\hat{v}_t} + \epsilon\big).
\end{aligned}$$

Initialization: $w_1 \in \mathbb{R}^d$ arbitrary, $m_0 = 0$, $v_0 = 0$, $\hat{v}_0 = 0$.

When we prove convergence below, we will focus on the deterministic-gradient special case or the momentum-free special case (explicitly stated).

Pseudocode

Inputs: w_1, $\{\eta_t\}_{t\geq 1}$, $\beta_1 \in [0,1)$, $\beta_2 \in [0,1)$, $\epsilon > 0$.

1. Set $m_0 \leftarrow 0$, $v_0 \leftarrow 0$, $\hat{v}_0 \leftarrow 0$.
2. For $t = 1, 2, \ldots, T$:
   (a) Compute stochastic gradient g_t at w_t.
   (b) Update first and second moments:
   $$m_t \leftarrow \beta_1 m_{t-1} + (1-\beta_1) g_t, \quad v_t \leftarrow \beta_2 v_{t-1} + (1-\beta_2)(g_t \odot g_t).$$
   (c) Bias-correct:
   $$\hat{m}_t \leftarrow \frac{m_t}{1-\beta_1^t}, \quad \tilde{v}_t \leftarrow \frac{v_t}{1-\beta_2^t}.$$
   (d) Monotone variance:
   $$\hat{v}_t \leftarrow \max(\hat{v}_{t-1},\, \tilde{v}_t).$$
   (e) Update parameters:
   $$w_{t+1} \leftarrow w_t - \eta_t\, \hat{m}_t \oslash \big(\sqrt{\hat{v}_t} + \epsilon\big).$$
3. Return w_{T+1} or an iterate selected by validation.

Dry Run Example

Target: $f(w) = (w-3)^2$ (scalar), gradient $\nabla f(w) = 2(w-3)$. Initialize $w_0 = 0$, choose $\eta_t \equiv \eta = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$.

Iter 1: $g_1 = 2(0-3) = -6$;

$$m_1 = 0.9 \cdot 0 + 0.1 \cdot (-6) = -0.6, \quad v_1 = 0.999 \cdot 0 + 0.001 \cdot 36 = 0.036;$$

$$\hat{m}_1 = \frac{-0.6}{1-0.9} = -6, \quad \tilde{v}_1 = \frac{0.036}{1-0.999} = 36, \quad \hat{v}_1 = \max(0, 36) = 36;$$

$$w_1 = 0 - 0.1 \cdot \frac{-6}{\sqrt{36} + 10^{-8}} \approx 0.1;$$

$$f(w_1) = (0.1-3)^2 = 8.41.$$

Iter 2: $g_2 = 2(0.1-3) = -5.8$;

$$m_2 = 0.9(-0.6) + 0.1(-5.8) = -1.12, \quad \hat{m}_2 = \frac{-1.12}{1-0.9^2} = \frac{-1.12}{0.19} \approx -5.89473684;$$

$$v_2 = 0.999 \cdot 0.036 + 0.001 \cdot 33.64 = 0.069604;$$

$$\tilde{v}_2 = \frac{0.069604}{1-0.999^2} = \frac{0.069604}{0.001999} \approx 34.81940970,$$

$$\hat{v}_2 = \max(36, 34.8194) = 36;$$

$$w_2 = 0.1 - 0.1 \cdot \frac{-5.89473684}{6 + 10^{-8}} \approx 0.19824561403;$$

$$f(w_2) \approx (0.1982456 - 3)^2 \approx 7.849811.$$

Iter 3: $g_3 = 2(0.1982456 - 3) \approx -5.60350877$;

$$m_3 = 0.9(-1.12) + 0.1(-5.60350877) \approx -1.56835088,$$

$$\hat{m}_3 = \frac{-1.56835088}{1-0.9^3} = \frac{-1.56835088}{0.271} \approx -5.787827;$$

$$v_3 = 0.999 \cdot 0.069604 + 0.001 \cdot 31.3953 \approx 0.100929696;$$

$$\tilde{v}_3 = \frac{0.100929696}{1-0.999^3} = \frac{0.100929696}{0.002997001} \approx 33.68,$$

$$\hat{v}_3 = \max(36, 33.68) = 36;$$

$$w_3 = 0.1982456 - 0.1 \cdot \frac{-5.787827}{6} \approx 0.2947094;$$

$$f(w_3) \approx (0.2947094 - 3)^2 \approx 7.322.$$

Assumptions

We analyze the momentum-free AMSGrad (set $\beta_1 = 0$, so $m_t = g_t$ and $\hat{m}_t = g_t$) under deterministic gradients $g_t = \nabla f(w_t)$.

- A1 (Smoothness): f is L-smooth: for all x, y, $f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y-x\|_2^2$.
- A2 (Lower bounded): $f_{\inf} := \inf_{x \in \mathbb{R}^d} f(x) > -\infty$.
- A3 (Bounded gradients in ℓ_∞): for all t and coordinates i, $|\nabla_i f(w_t)| \le G_\infty$.
- A4 (Stepsizes): $\eta_t = \eta/\sqrt{t}$ with $0 < \eta \le \epsilon/(2L)$.

Remark: A3 implies $\tilde{v}_{t,i} \le G_\infty^2$ and, hence, $\hat{v}_{t,i} \le G_\infty^2$ for all t.

Main Convergence Theorem

Theorem 1 (Non-convex rate for momentum-free AMSGrad)**.** *Under A1–A4, for any $T \geq 1$, the iterates of AMSGrad with $\beta_1 = 0$ satisfy*

$$\min_{1 \leq t \leq T} \|\nabla f(w_t)\|_2^2 \;\leq\; \frac{(\epsilon + G_\infty)^2}{\eta\,(\sqrt{T} - 1)}\,\big(f(w_1) - f_{\inf}\big).$$

Equivalently, $\min_{1\leq t\leq T} \|\nabla f(w_t)\|_2^2 = \mathcal{O}\Big(\frac{1}{\sqrt{T}}\Big)$ *with explicit constant* $\frac{(\epsilon+G_\infty)^2}{\eta}$ *times the initial sub-optimality.*

Theoretical Properties

- Stability via monotone preconditioner: $\hat{v}_t$ is elementwise nondecreasing, so the effective per-coordinate stepsizes $\eta_t/(\sqrt{\hat{v}_{t,i}} + \epsilon)$ are nonincreasing, preventing exploding updates.
- Hyperparameter sensitivity: - Larger ϵ increases the lower bound on the denominator, reducing stepsizes but improving the constant in the descent sufficient condition $\eta \leq \epsilon/(2L)$. - Larger β_2 makes $\tilde{v}_t$ smoother; monotone max prevents transient decreases in the denominator.
- Comparison to SGD: In the momentum-free, deterministic setting, the rate matches the canonical non-convex SGD rate $O(1/\sqrt{T})$, while AMSGrad adaptively rescales coordinates to mitigate ill-conditioning.

Discussion

- The proof below is for $\beta_1 = 0$ (no momentum). Extending to $\beta_1 > 0$ adds bias-correlation terms between $\hat{m}_t$ and $\hat{v}_t$; with additional technical conditions one obtains the same $O(1/\sqrt{T})$ rate up to constants. - In the stochastic setting, one typically obtains an $O(1/\sqrt{T})$ rate for the expected gradient norm under standard unbiasedness/variance assumptions, with constants depending on σ, ϵ, and G_∞.

Preliminaries (Definitions and Lemmas)

Throughout, all inequalities hold elementwise when applied to vectors unless otherwise stated. Define the diagonal matrix

$$H_t := \mathrm{diag}\big(\sqrt{\hat{v}_t} + \epsilon\big) \in \mathbb{R}^{d\times d}, \qquad D_t := H_t^{-1}.$$

Thus the update is $w_{t+1} = w_t - \eta_t D_t \nabla f(w_t)$ in the momentum-free deterministic setting.

Lemma 1 (Smoothness one-step bound)**.** *For any $t \geq 1$, with $y = w_{t+1}$ and $x = w_t$,*

$$f(w_{t+1}) \leq f(w_t) + \langle \nabla f(w_t), w_{t+1} - w_t\rangle + \frac{L}{2}\|w_{t+1} - w_t\|_2^2.$$

Proof. *This is the standard descent lemma for L-smooth functions. [Step 1]*

Lemma 2 (Preconditioner bounds)**.** *Under A3, for all t,*

$$\epsilon I \ \preceq\ H_t \ \preceq\ (\epsilon + G_\infty)\, I.$$

Proof. *Because $|g_{t,i}| \le G_\infty$ (A3), we have $g_{t,i}^2 \le G_\infty^2$, hence $v_{t,i} \le G_\infty^2$, so $\tilde v_{t,i} \le G_\infty^2$ and $\hat v_{t,i} \le G_\infty^2$ by monotonicity of the max. Therefore $0 \le \sqrt{\hat v_{t,i}} \le G_\infty$ and $\epsilon \le \sqrt{\hat v_{t,i}} + \epsilon \le \epsilon + G_\infty$ for each i. This yields the claimed matrix inequality. [Step 2]*

Lemma 3 (One-step descent inequality (deterministic, $\beta_1 = 0$))**.** *Under A1 and with $w_{t+1} = w_t - \eta_t D_t \nabla f(w_t)$,*

$$f(w_{t+1}) \le f(w_t) - \eta_t \|\nabla f(w_t)\|_{D_t}^2 + \frac{L}{2}\eta_t^2 \|\nabla f(w_t)\|_{D_t^2}^2.$$

Proof. *Set $\Delta_t := w_{t+1} - w_t = -\eta_t D_t \nabla f(w_t)$. Apply Lemma 1:*

$$\begin{aligned}
f(w_{t+1}) &\le f(w_t) + \langle \nabla f(w_t), \Delta_t\rangle + \frac{L}{2}\|\Delta_t\|_2^2 \quad \textit{(by smoothness)} \quad \textit{[Step 3]}\\
&= f(w_t) - \eta_t \langle \nabla f(w_t), D_t \nabla f(w_t)\rangle + \frac{L}{2}\eta_t^2 \|D_t \nabla f(w_t)\|_2^2 \quad \textit{[Step 4]}\\
&= f(w_t) - \eta_t \|\nabla f(w_t)\|_{D_t}^2 + \frac{L}{2}\eta_t^2 \|\nabla f(w_t)\|_{D_t^2}^2, \quad \textit{[Step 5]}
\end{aligned}$$

where $\|x\|_D := \sqrt{x^\top D x}$; since D_t is diagonal and positive definite, $\|D_t x\|_2^2 = x^\top D_t^\top D_t x = x^\top D_t^2 x = \|x\|_{D_t^2}^2$.

Lemma 4 (Relating D_t^2 and D_t)**.** *For all t, $\|\nabla f(w_t)\|_{D_t^2}^2 \le \frac{1}{\epsilon}\, \|\nabla f(w_t)\|_{D_t}^2$.* ***Proof.*** *Since $H_{t,ii} \ge \epsilon$ for all i (Lemma 2), we have $D_{t,ii} = 1/H_{t,ii} \le 1/\epsilon$. Therefore $D_{t,ii}^2 \le \frac{1}{\epsilon} D_{t,ii}$ for all i. Summing coordinate-wise yields the inequality. [Step 6]*

Lemma 5 (Sums for the stepsize schedule)**.** *Let $\eta_t = \eta/\sqrt{t}$ with $\eta > 0$. Then, for $T \ge 1$,*

$$\sum_{t=1}^{T} \eta_t \ \ge\ 2\eta(\sqrt{T} - 1), \qquad \sum_{t=1}^{T} \eta_t^2 \ \le\ \eta^2\,(1 + \log T).$$

Proof. *For the first bound, use the integral comparison $\sum_{t=1}^{T} \frac{1}{\sqrt{t}} \ge \int_1^T \frac{1}{\sqrt{x}}\, dx = 2(\sqrt{T} - 1)$ and multiply by η. [Step 7] For the second, $\sum_{t=1}^{T} \frac{1}{t} \le 1 + \log T$; multiply by η^2. [Step 8]*

Main Proof (Step-by-Step Derivation)

We prove the theorem under A1–A4 with $\beta_1 = 0$. From Lemma 3 and Lemma 4, for all $t \ge 1$,

$$\begin{aligned}
f(w_{t+1}) &\le f(w_t) - \eta_t \|\nabla f(w_t)\|_{D_t}^2 + \frac{L}{2}\eta_t^2 \|\nabla f(w_t)\|_{D_t^2}^2 \quad \text{[Step 9]}\\
&\le f(w_t) - \eta_t \|\nabla f(w_t)\|_{D_t}^2 + \frac{L}{2}\eta_t^2 \cdot \frac{1}{\epsilon}\, \|\nabla f(w_t)\|_{D_t}^2 \quad \text{(by Lemma 4)} \quad \text{[Step 10]}\\
&= f(w_t) - \Big(\eta_t - \frac{L}{2\epsilon}\eta_t^2\Big)\, \|\nabla f(w_t)\|_{D_t}^2. \quad \text{[Step 11]}
\end{aligned}$$

Define the (nonnegative) coefficient

$$\gamma_t := \eta_t - \frac{L}{2\epsilon}\eta_t^2 = \frac{\eta_t}{2}\Big(2 - \frac{L}{\epsilon}\eta_t\Big). \quad \text{[Step 12]}$$

By A4, $\eta_t \le \eta \le \epsilon/(2L)$, hence $2 - \frac{L}{\epsilon}\eta_t \ge 2 - \frac{L}{\epsilon}\eta \ge 1$, so $\gamma_t \ge \frac{\eta_t}{2}$. [Step 13] Thus

$$f(w_{t+1}) \le f(w_t) - \gamma_t \|\nabla f(w_t)\|_{D_t}^2 \le f(w_t) - \frac{\eta_t}{2}\, \|\nabla f(w_t)\|_{D_t}^2. \quad \text{[Step 14]}$$

Next, lower bound the preconditioned norm using Lemma 2:

$$\|\nabla f(w_t)\|_{D_t}^2 = \nabla f(w_t)^\top H_t^{-1} \nabla f(w_t) \ge \frac{1}{(\epsilon + G_\infty)^2}\, \|\nabla f(w_t)\|_2^2. \quad \text{[Step 15]}$$

Plugging this into [Step 14],

$$f(w_{t+1}) \le f(w_t) - \frac{\eta_t}{2(\epsilon + G_\infty)^2}\, \|\nabla f(w_t)\|_2^2. \quad \text{[Step 16]}$$

Summing [Step 16] over $t = 1$ to T gives the telescoping bound

$$f(w_{T+1}) \le f(w_1) - \frac{1}{2(\epsilon + G_\infty)^2} \sum_{t=1}^{T} \eta_t\, \|\nabla f(w_t)\|_2^2. \quad \text{[Step 17]}$$

Using A2 ($f(w_{T+1}) \ge f_{\inf}$), we obtain

$$\sum_{t=1}^{T} \eta_t\, \|\nabla f(w_t)\|_2^2 \le 2(\epsilon + G_\infty)^2 \big(f(w_1) - f_{\inf}\big). \quad \text{[Step 18]}$$

Let $t^\star \in \arg\min_{1 \le t \le T} \|\nabla f(w_t)\|_2^2$. Then

$$\|\nabla f(w_{t^\star})\|_2^2 \cdot \sum_{t=1}^{T} \eta_t \;\le\; \sum_{t=1}^{T} \eta_t\, \|\nabla f(w_t)\|_2^2 \;\le\; 2(\epsilon + G_\infty)^2 \big(f(w_1) - f_{\inf}\big). \quad \text{[Step 19]}$$

Using Lemma 5, $\sum_{t=1}^{T} \eta_t \ge 2\eta(\sqrt{T} - 1)$, hence

$$\min_{1 \le t \le T} \|\nabla f(w_t)\|_2^2 \;=\; \|\nabla f(w_{t^\star})\|_2^2 \;\le\; \frac{2(\epsilon + G_\infty)^2 \big(f(w_1) - f_{\inf}\big)}{2\eta(\sqrt{T} - 1)} \;=\; \frac{(\epsilon + G_\infty)^2}{\eta(\sqrt{T} - 1)} \big(f(w_1) - f_{\inf}\big). \quad \text{[Step 20]}$$

This proves the theorem. □

Rate Derivation (With Constants)

From [Step 20],

$$\min_{1 \le t \le T} \|\nabla f(w_t)\|_2^2 \le \frac{(\epsilon + G_\infty)^2}{\eta} \cdot \frac{f(w_1) - f_{\inf}}{\sqrt{T} - 1}.$$

For $T \ge 4$, $\sqrt{T} - 1 \ge \frac{1}{2}\sqrt{T}$, hence the simpler bound

$$\min_{1 \le t \le T} \|\nabla f(w_t)\|_2^2 \le \frac{2(\epsilon + G_\infty)^2}{\eta} \cdot \frac{f(w_1) - f_{\inf}}{\sqrt{T}}.$$

Moreover, averaging the gradients yields

$$\frac{\sum_{t=1}^{T} \eta_t \|\nabla f(w_t)\|_2^2}{\sum_{t=1}^{T} \eta_t} \le \frac{2(\epsilon + G_\infty)^2 \big(f(w_1) - f_{\inf}\big)}{2\eta(\sqrt{T} - 1)} = \frac{(\epsilon + G_\infty)^2 \big(f(w_1) - f_{\inf}\big)}{\eta(\sqrt{T} - 1)}.$$

Special Cases

- Constant stepsize: If $\eta_t \equiv \eta \le \epsilon/(2L)$, then [Step 18] becomes $\sum_{t=1}^{T} \|\nabla f(w_t)\|_2^2 \le \frac{2(\epsilon+G_\infty)^2}{\eta}\big(f(w_1) - f_{\inf}\big)$, so $\min_{1\le t\le T} \|\nabla f(w_t)\|_2^2 \le \frac{2(\epsilon+G_\infty)^2}{\eta} \cdot \frac{f(w_1)-f_{\inf}}{T}$.

- With momentum ($\beta_1 > 0$): Writing $\hat{m}_t$ as a convex combination of past gradients introduces additional bias terms coupling $\hat{m}_t$ and H_t. Under standard bounded-gradient and smoothness assumptions, and with $\eta_t = \eta/\sqrt{t}$, one can extend the proof by (i) lower bounding $\langle \nabla f(w_t), \hat{m}_t \rangle$ using Lipschitz continuity to control drift across iterations, and (ii) reusing Lemmas 2 and 4 to control quadratic terms. The same $O(1/\sqrt{T})$ rate holds up to constants depending on β_1.

- Stochastic gradients: If g_t is an unbiased estimator of $\nabla f(w_t)$ with bounded variance, a similar proof goes through with expectations; the noise contributes additional terms that remain controlled under the decreasing stepsize $\eta_t = \eta/\sqrt{t}$, yielding an $O(1/\sqrt{T})$ rate in expectation for the minimal gradient norm.